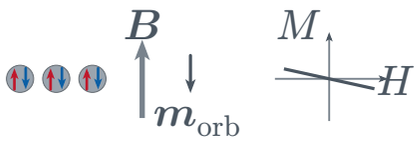
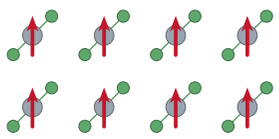
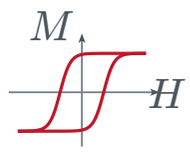
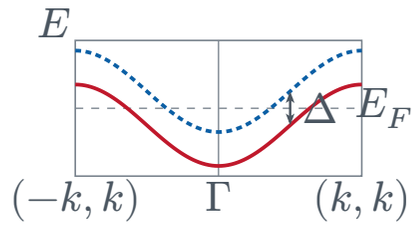
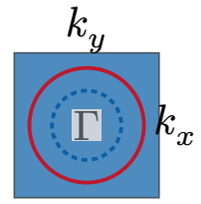
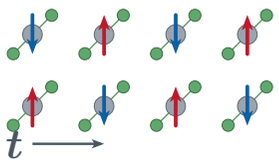
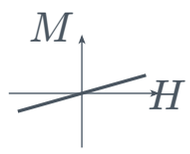
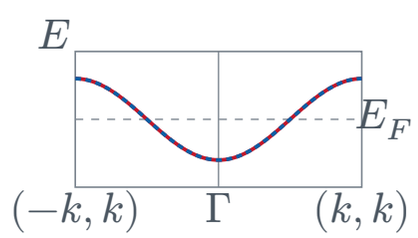
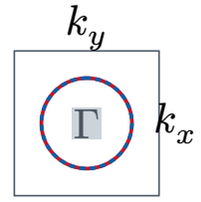
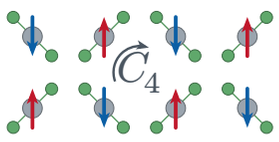
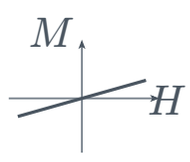
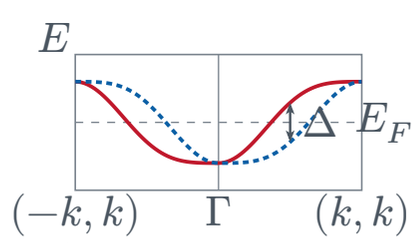
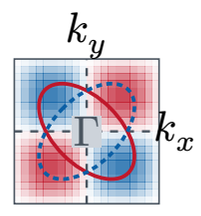
all three rows are collinear; $M(H)$ singles out the ferromagnet, and only the spin splitting $\Delta(\boldsymbol{k}) = E_\uparrow(\boldsymbol{k}) - E_\downarrow(\boldsymbol{k})$ separates the other two

| | **real-space structure** | **response $M(H)$** | **spin-resolved bands** | **spin splitting $\Delta(\boldsymbol{k})$** |
|---|---|---|---|---|
| **Diamagnetism** |  closed shells, $\boldsymbol{m}_{\text{orb}}$ opposes $\boldsymbol{B}$ $\chi \sim -10^{-5} < 0$ | | | |
| **Ferromagnetism** | one sublattice, every moment parallel |  identical ligand axes |  hysteretic, $M_r \neq 0$ |  rigid $s$-wave split, same sign at every $\boldsymbol{k}$ |  concentric contours, majority larger |
| **Paramagnetism** |  disordered moments, $\langle \boldsymbol{m} \rangle = 0$ $\chi = C/T \sim 10^{-3} > 0$ | | | |
| **Antiferromagnetism** | sublattices related by a translation $\boldsymbol{t}$, or by inversion |  identical ligand axes |  linear, no remanence |  $E_\uparrow(\boldsymbol{k}) = E_\downarrow(\boldsymbol{k})$ everywhere |  one doubly degenerate contour |
| **Ferrimagnetism** |  antiparallel but $\lvert\boldsymbol{m}_A\rvert \neq \lvert\boldsymbol{m}_B\rvert$, so $\lvert\boldsymbol{M}_A + \boldsymbol{M}_B\rvert \neq 0$ | | | |
| **Altermagnetism** | sublattices related only by a $C_4$ rotation |  ligand axes rotated 90° |  identical to the antiferromagnet |  $\Delta$ flips between the $C_4$-related diagonals |  $\Delta \propto \sin k_x \sin k_y$, nodal on axes |

$E_\uparrow$ $E_\downarrow$ curves drawn on top of each other are spin degenerate $\Delta > 0$ $\Delta < 0$

non-relativistic limit throughout; $M(H)$ axes not to scale across panels; $d$-wave altermagnet shown, $g$- and $i$-wave also exist